

Wyatt Madden M.S.

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Department of Biostatistics & Bioinformatics
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EDUCATION	Emory University Ph.D. in Biostatistics & Bioinformatics 2021 – Present
	Montana State University M.S. in Statistics 2017 – 2019
	University of California, Santa Cruz Bachelor of Arts, Economics & Mathematics <i>With Honors</i> Bachelor of Arts, Film & Digital Media <i>With Honors</i> 2011 – 2015
RESEARCH INTERESTS	Bayesian computation, spatio-temporal statistics, probabilistic machine learning & deep learning, with applications in environmental health and infectious disease.
PUBLICATIONS	1. W. G. Madden, Y. Liu, and H. H. Chang, “Scalable bayesian geostatistical regression model for bias-correcting large-scale daily satellite-retrieved aerosol optical depth and chemical transport model simulations,” <i>Atmospheric Environment</i>, vol. 380, p. 122 079, 2026, Full Paper. 2. J. H. Lagergren, M. Ruiz-Aravena, D. J. Becker, <i>et al.</i>, “Environmental and ecological signals predict food shortages for subtropical populations of australian flying foxes, reservoirs of hendra virus,” <i>Biology Letters</i>, vol. 21, no. 12, p. 20 250 371, Dec. 2025, Full Paper. 3. W. G. Madden, M. Qi, Y. Liu, and H. H. Chang, “Ensembledownscaler: R package for bayesian ensemble averaging of pm2.5 geostatistical downscalers,” <i>Remote Sensing</i>, vol. 17, no. 11, 2025, Full Paper. 4. W. G. Madden, W. Jin, B. Lopman, A. Zufle, B. Dalziel, C. J. E. Metcalf, B. T. Grenfell, and M. S. Y. Lau, “Deep neural networks for endemic measles dynamics: Comparative analysis and integration with mechanistic models,” <i>PLoS Computational Biology</i>, vol. 20, no. 11, e1012616, Nov. 2024, Full Paper. 5. P. Eby, A. Peel, A. Hoegh, W. Madden, J. Giles, P. Hudson, and R. Plowright, “Pathogen spillover driven by rapid changes in bat ecology,” <i>Nature</i>, pp. 1–3, Nov. 2022, Full Paper. 6. D. J. Becker, P. Eby, W. Madden, A. J. Peel, and R. K. Plowright, “Ecological conditions predict the intensity of hendra virus excretion over space and time from bat reservoir hosts,” <i>Ecology Letters</i>, Oct. 2022, Full Paper. 7. M. S. Y. Lau, A. Becker, W. Madden, L. A. Waller, C. J. E. Metcalf, and B. T. Grenfell, “Comparing and linking machine learning and semi-mechanistic models for the predictability of endemic measles dynamics,” <i>PLOS Computational Biology</i>, vol. 18, no. 9, pp. 1–14, Sep. 2022, Full Paper. 8. M. D. Cherne, A. B. Gentry, A. Nemudraia, <i>et al.</i>, “Severe acute respiratory syndrome coronavirus 2 is detected in the gastrointestinal tract of asymptomatic endoscopy patients but is unlikely to pose a significant risk to healthcare personnel,” <i>Gastro Hep Advances</i>, vol. 1, no. 5, pp. 844–852, 2022, Full Paper. 9. A. Hoegh, A. Peel, W. Madden, M. Ruiz-Aravena, A. Morris, A. Washburne, and R. Plowright, “Estimating viral prevalence with data fusion for adaptive two-phase pooled sampling,” <i>Ecology and Evolution</i>, vol. 11, Sep. 2021, Full Paper. 10. W. Rogers, M. Ruiz-Aravena, D. Hansen, <i>et al.</i>, “High-frequency screening combined with diagnostic testing for control of sars-cov-2 in high-density settings: An economic evaluation of resources allocation for public health benefit,” <i>medRxiv</i>, 2021, Under Review.
GRANT SUPPORT	As Graduate Student/Trainee (✓ Completed) <i>Graduate and Post-doctoral Training in Environmental Health Science and Toxicology Training Grant</i> Source: NIEHS T32 ES012870 PI: Carmen Marsit Direct Cost: \$28,788 2025–2026

INVITED PRESENTATIONS	Mechanism-Integrated Machine Learning for Infectious Disease Dynamics Prediction High Meadows Environmental Institute Workshop On the Accuracy (and Niceness) of Prediction: from Epidemics to Climate and Weather Princeton, NJ	Oct 2024
	Machine Learning Approaches for Epidemic Modeling Princeton Serology Conference Princeton, NJ	Mar 2023
	Compartmental Models: Deterministic & Bayesian Approaches Rocky Mountain Data Science Bozeman, MT	Nov 2020
	R Studio in Action - DataFest Montana ASA Chapter Meeting Bozeman, MT	Oct 2018
CONTRIBUTED TALKS & POSTERS	Bayesian Ensemble with Spatially Varying Covariate-Dependent Weights for PM _{2.5} Estimation ISEE-NAC 2026 [Oral Presentation] Baltimore, USA	Jun 2026
	Scalable Bayesian Geostatistical Regression Model for Bias-Correcting Large-Scale Daily Satellite-Retrieved AOD and CTM Simulations ISEE 2025 [Poster] Atlanta, USA	Aug 2025
	Neural Network Reveals Gravitational Coupling of Endemic Measles Dynamics Epidemics9 [Poster] Bologna, Italy	Dec 2023
	Bias-Correcting Daily Satellite-Retrieved AOD for Air Quality Research EnviBayes Workshop [Poster] Fort Collins, CO	Sep 2023
INVITED PANELS	Machine Learning Panel CIDMATH Retreat Atlanta, GA	Mar 2024
PROFESSIONAL EXPERIENCE	Los Alamos National Laboratory Applied Machine Learning Research Fellow - Developed deep learning methods for high energy density experiments. - Designed and implemented PyTorch model fitting pipelines for use on high performance computing clusters.	Los Alamos, NM May 2024 – Aug 2024
	Bozeman Disease Ecology Lab Statistician - Researched spatio-temporal data integration techniques for viral surveillance and prediction. - Provided statistics & machine learning consulting for international team of scientists. - Developed R packages to automate routine statistical analysis, visualization, and wrangling. - Designed and implemented SQL database and data pipelines, ensuring data quality and access.	Bozeman, MT Jan 2019 – Jul 2021
	Weyerhaeuser Statistics Intern	Seattle, WA May 2018 – Aug 2018

- Implemented machine learning models aimed at lowering defects in industrial processes, after diagnosing issues through exploratory visualization and analyses.
- Formulated mixed-model experimental designs.
- Developed Shiny web applications to automate data cleaning/wrangling workflows.

Accenture

Analyst

Sacramento, CA
Jul 2016 – Apr 2017

- Improved loan approval processes through analysis of credit profiles.

CONSULTING & COLLABORATION EXPERIENCE

Collaborator

Aug 2018 – Dec 2018

Statistical Consulting And Research Services (SCRS)
Department of Mathematical Sciences, Montana State University

Volunteer

Jan 2018 – Apr 2018

Statistics Without Borders (SWB)
Under direction of Dr. Nicole Carnegie, Montana State University

TEACHING

Instructor, Creator

Fall 2025

Nearest-Neighbor Gaussian Processes for Large Spatial Datasets
Department of Biostatistics and Bioinformatics, Emory University

Instructor, Creator

Spring 2023 – Spring 2025

Neural Networks with PyTorch Tutorial
Department of Biostatistics and Bioinformatics, Emory University

Teaching Assistant

Fall 2022 – Spring 2025

INFO 534 - Applied Machine Learning
Department of Biostatistics and Bioinformatics, Emory University

Teaching Assistant

Summer 2024

Introduction to Machine Learning for ID Modeling
Summer Institute in Statistics and Modeling in Infectious Diseases, Emory University

Instructor

Fall 2017 – Fall 2018

MATH 105 - Contemporary Mathematics
Department of Mathematical Sciences, Montana State University

AWARDS

Patel-Naik Award (2nd Place), Emory University

Dec 2023

Outstanding Graduate Student Award, Montana State University

May 2019

Excellence in Data Visualization, ASA Data Fest - Montana State University

Apr 2018

SERVICE

Emory BIOS Student Council, Pre-quals Representative

Spring 2022 – Present

Georgia Statistics Day 2024, Student Volunteer

Oct 25th, 2024

Georgia Statistics Day 2021, Student Volunteer

Oct 11th, 2021

Bozeman Environmental Statistics Group, Member

2019 – 2021

American Statistical Association Student Chapter at Montana State, Treasurer

2018 – 2019

MEMBERSHIP

American Statistical Association